Erdős #203: literature ledger, audit and untried methods

Prepared for Deepnil Ray – branch `claude/open-conjecture-search-h6tgg3`, folder `e203/`

30 October 2026

Problem. Is there m with $(m,6)=1$ such that $2^k3^\ell m+1$ is composite for all k,ℓ (site form $k,\ell\geq 0$; the repository works with $k,\ell\geq 1$, which is the weaker YES statement and the stronger NO statement, so every repository exclusion of YES and every repository NO-witness transfers to the site form)?

Status. Open. Nothing in this document proves either answer. Labels used throughout: PROVED, VERIFIED (computation with a control that can fail), CONJECTURED, and for citations P (page opened this session or recorded as opened in an earlier ledger), S (second-hand), U (not opened).

1 Audit of the existing write-up

Re-run today with code independent of the repository's builders. Every check below includes a control that was shown able to fail.

Claim	Result of the re-run	Status
N1: hardest $m\leq 10^{10}$ is 6257518159, first prime at $(16,10)$	$2^{16}3^{10}m+1$ is prime and is the smallest-size prime cell with $k,\ell\geq 1$ (next cells $(29,2)$, $(26,4)$). $\gcd(m,6)=1$. Control $m=1$ returns $(1,1)$, i.e. 7.	CONFIRMED
Y4: the pool of primes with $e_p\leq 8\cdot 10^5$ is complete	Orders of 2 and 3 recomputed from scratch for every prime $5\leq p<3\cdot 10^6$ (spf sieve, exact order test). 88 923 primes have $e_p\leq 8\cdot 10^5$ and equal the pool restricted to $p<3\cdot 10^6$ exactly. Deleting one pool prime makes the comparison fail (control fired).	CONFIRMED for $p<3\cdot 10^6$ (the repository itself reaches $2\cdot 10^9$)
Pool lattice forms (α,β,e)	15 000 random (k,ℓ) on 3 000 random pool primes: $\alpha k+\beta\ell\equiv 0 \pmod e$ iff $2^k3^\ell\equiv 1 \pmod p$, zero mismatches.	CONFIRMED (sample)
Multiplicities $c(e)$	Histogram {1:89200, 2:5412, 3:250, 4:10}, $\sum c(e)^2=113\,258$, $\sum c(e)=100\,814$. Matches L21.	CONFIRMED
Y7: primes 5 and 13 are incompatible in a common-T covering	2 generates $(\mathbb{Z}/5)^\times$ and $3=2^3$, forcing $T\equiv 3 \pmod 4$; 2 generates $(\mathbb{Z}/13)^\times$ and $3=2^4$, forcing $T\equiv 0 \pmod 4$. Incompatible.	CONFIRMED (by hand)
Citations	Eleven arXiv identifiers opened (section 3). Authors, titles and dates match, with the corrections listed at the end of section 3.	CONFIRMED with corrections

Claim	Result of the re-run	Status
Numeric odds in repository documents	`VERDICT.md` and `verdict/1_heuristic.md` stated "about 95%", `TARGET.md` stated "about 10%". Removed today, in line with the standing rule against success probabilities.	FIXED
HUB run state	HUB lists three exact verifiers as RUNNING. The container was replaced, so none was running. Checkpoints exist (58, 103 and 6 lines) and none has finished; `C_verify_final.log` is empty. Two were resumed today in the background.	CORRECTED
Y1 bound 0.730584, Theorem Z, Theorem A sums	Not re-run (hours of exact-rational arithmetic, or an interval-arithmetic rewrite). Agent V's red-team log confirms the pieces listed in `V_check.md`.	NOT RE-RUN
Theorem A input (Friedlander–Iwaniec, *Opera de Cribro*, Lemma 6.3, error $e^{9\kappa-s}K^{10}$)	Searched again. Wikipedia and the Columbia sieve-seminar notes do not quote the lemma. The explicit form is still unconfirmed by any page I opened.	STILL OPEN

2 Level of what closing #203 requires

The ladder below places each rung on the significance scale anchored at 100 = Riemann hypothesis. The placements are my judgment and carry no measurement behind them.

Rung	Target	Rough scale	Status
0	Finite verification: every $m \le 10^{10}$ has a prime cell	—	done (N1)
1	Paper: Theorems Y1, Y5 (Z), N3 (A), lemmas Y7, N5; exact-rational checks; Opera de Cribro match	about 8–12	four items open, all closable
2	"No covering exists at any size" (kills every known YES mechanism)	about 18–25	needs a bound on $c(e) = \#\{p : e_p = e\}$; weaker than Erdős's one-base conjecture $E_a(r) \ll r^{\varepsilon}$, still unproved
3	Full YES: explicit covering or mixed certificate	about 25–30	must use at least 14 082 primes with $e_p > 8 \cdot 10^5$; no construction method known

Rung	Target	Rough scale	Status
4	Full NO: a prime for every m	about 35 and above	a family with $O((\log x)^2)$ members per m and the parity barrier; Erdős–Odlyzko gives positive lower density only

Closing #203 itself sits above the 5–19 band. Rung 1 lies inside it. Rung 2 lies just above it and is the nearest honest target for a result that moves the problem.

3 Consolidated literature ledger

Entries come from `APPROACHES.md`, `L_literature.md`, the public pages, and this session's searches. The last column is the verification flag.

3.1 Origin and public state

#	Source	Content and bearing	Flag
1	Erdős–Graham, *Old and new problems and results in combinatorial number theory*, Monogr. Enseign. Math. 28 (1980), p. 27	Origin. Tao confirmed the statement on the forum, 29 Dec 2025.	S
2	Erdős–Odlyzko, J. Number Theory 11 (1979) 257–263	Positive lower density of m with a prime cell (Thm 2, any prime set). Moment method on the exponent box, Linnik-type lower bound, sieve upper bound, Romanoff-type order sums. Best known toward NO; no explicit constant.	P
3	erdosproblems.com/203	Open, "cannot be resolved with a finite computation", last edited 20 Jan 2026, formalised, user uona working, no claimed solutions. Re-read today, unchanged.	P
4	erdosproblems.com/1113	One-base cousin: Sierpiński numbers without a covering set. Open, last edited 29 Dec 2025, no two-base content. Re-read today.	P
5	Forum posts (AnimishSharma Jun 2026; emil467q Nov 2025; veljjanoski Sep 2026)	$\sum 1/\lvert H_p\rvert \geq 1$ criterion and no covering with lcm < 5040 for $p \leq 5 \cdot 10^7$; no $m \in [10^{10}, 10^{11}]$; coverage plateaus 71.7% at $N = 5040$, 76.0% at $N = 55440$.	P
6	github.com/Neo7672/erdos-203-covering-search (16–17 Sep 2026, AI-assisted, 7 commits)	*New this session.* 105 primes below $5 \cdot 10^6$ with $h(p) \leq 400$, total density 1.63; budget 1.02 at $N = 5040$; search plateaus with 17–21% uncovered. Claims no solution. Far weaker than Y1.	P

#	Source	Content and bearing	Flag
7	Issue #9414, the-omega-institute/trureturing	*New this session.* Progress log, states "whole-solution KPI: 0". Reports that a fixed 252-row family leaves at least 1.48676% uncovered under any global phase vector. No covering or proof claimed.	P (page); claims unchecked
8	veljjanoski/erdos203 (GitHub)	Negative computational search, no proof content.	P

3.2 One-base covering theory and constructions

#	Source	Content and bearing	Flag
9	Sierpiński, Elem. Math. 15 (1960); Selfridge (1962/63); Robinson, Proc. AMS 9 (1958)	Coverings of $\{k2^n+1\}$ by primes. No two-dimensional analogue known.	P (bibliography)
10	Izotov, Fibonacci Quart. 33 (1995); Filaseta–Finch–Kozek, J. Number Theory 128 (2008) 1916–1940	Covering plus algebraic factors (Sophie Germain). Conjecture: a Sierpiński number is a perfect power or has bounded least prime factor. Handled in `MIXED.md` via Capelli.	P
11	Simpson (1985); Jenkin–Simpson, INTEGERS 3 (2003) A13	Irredundant 1-D covering with lcm $\prod p_i^{a_i}$ has at least $1+\sum a_i(p_i-1)$ classes.	P
12	Filaseta–Groth–Luckner arXiv:2305.09219; Filaseta–Juillerat–Luckner arXiv:2209.10646; Grantham et al. arXiv:1903.05023; Agrawal et al. arXiv:2208.09720; Harrington–Sun–Wong arXiv:2104.00602	Constructions and classifications, all with one exponent and a free multiplier fixed by CRT. Not usable here because the lattices Λ_p are fixed by p.	P

3.3 Minimum modulus and the distortion method (YES-side obstruction)

#	Source	Content and bearing	Flag
13	Hough, Ann. of Math. 181 (2015) 361–382	Least modulus $\le 10^{16}$ (Annals), 10^{18} on the arXiv page. Cite the Annals value.	P
14	Balister–Bollobás–Morris–Sahasrabudhe–Tiba, Invent. Math. 228 (2022) 377–414	Distortion method; engine of `DISTORTION.md`.	P
15	Hough–Nielsen, Proc. AMS (2019), arXiv:1703.02133	Distinct moduli: some modulus divisible by 2 or 3. Repeated moduli not treated.	P

#	Source	Content and bearing	Flag
16	Klein–Koukoulopoulos–Lemieux, arXiv:2212.01299 (v2, Aug 2023)	j-th smallest modulus $\le \exp(cj^2/\log(j+1))$; multiplicity-s version $\exp(c\log^2(s+1)/\log\log(s+2))$ as reported in L5.	P (abstract); multiplicity theorem S
17	Li–Wang–Yi, arXiv:2302.05946 (v2, Aug 2026)	Distortion method as an abstract theorem on a finite set with an inverse system; the repository's tilt is a special case (V confirmed).	P

3.4 Lattice and coset coverings

#	Source	Content and bearing	Flag
18	Granville–Pappalardi, arXiv:2601.10296 (v2, Apr 2026)	Two-dimensional lattice coverings for $a^m - b^n$; Conjecture 1 that the only obstruction is a covering. Does not mention #203. Item 5 there proves non-existence for $a = 2$ by a homogeneity point that has no coset analogue.	P
19	Cremona–Koymans, arXiv:2601.03212 (v3, 12 Oct 2026)	Finite coverings of $\mathbb{Z}^2$ by sublattices (homogeneous only). Size bound $t + 1 + \sum e_i(p_i - 1)$. Does not transfer to cosets.	P
20	Lettl–Sun, arXiv:math/0411144 (v2, 2008)	For a minimal cover by cosets, every irredundant coset has index $\le 2^{k-m}$ and $k \ge m + f(\text{index})$ with f the Mycielski function. Used for pruning; weak.	P
21	Snellman, arXiv:2608.22550 (23 Aug 2026)	Dense directions in lattice-line coverings of $\mathbb{Z}^2$. No bearing.	P
22	Borodzik–Nguyen–Robins, arXiv:1306.2644	Exact tilings by translated sublattices. Certificates overlap, so no use.	S
23	Hegedüs, arXiv:2405.16231 (v2, Apr 2026); Jamison; Brouwer–Schrijver	Almost covers and blocking sets in affine space; quantitative support for the direction lemma.	P (abstract)

3.5 Orders, gcds and the pool

#	Source	Content and bearing	Flag
24	Bugeaud–Corvaja–Zannier, Math. Z. 243 (2003) 79–84	$\gcd(a^n - 1, b^n - 1) < e^{\varepsilon n}$, ineffective. Gives $c(e) \le \varepsilon e/\log e$ only.	P
25	Corvaja–Zannier, arXiv:math/0311030; Hernandez–Luca	Common factors of shifted S-units, finitely many exceptional pairs, ineffective.	S

#	Source	Content and bearing	Flag
26	Nguyen-Dang, arXiv:2606.07959 (v2, Aug 2026)	$\gcd(a^n-1, b^n-1)$ satisfies a linear recurrence only for dependent a, b.	P
27	Akbary–Yazdani, arXiv:1505.06500	States Erdős's conjecture $E_a(r) \ll r^{\varepsilon}$; abstract also covers function fields and elliptic curves. The conjecture statement was read in an earlier pass, not re-read today.	P (abstract); statement S
28	Pehlivan, arXiv:1504.01550 (DOI 10.1142/S1793042116501281)	Lemma 8: $\#\{p : \lvert\Gamma_p\rvert \le t\} \ll t^{1+1/r}/\log t$. The abstract mentions only GRH; the unconditional status of Lemma 8 was read earlier and is unconfirmed today.	P (abstract); lemma S
29	Matthews (1982); Erdős–Murty (1999); Elsholtz, Proc. AMS 130 (2002); Kurlberg–Pomerance, Acta Arith. 119 (2005)	Almost-all-p lower bounds on $\lvert\langle 2,3\rangle \bmod p\rvert$. Nothing on the exceptional pool.	S/P
30	Zsigmondy; Bilu–Hanrot–Voutier (2001); Evertse (1984)	Primitive divisors; finiteness of S-unit solutions. Structural only.	P

3.6 NO-side tools

#	Source	Content and bearing	Flag
31	Heath-Brown (1992); Xylouris, arXiv:0906.2749	Linnik constant 5.2; gives m for a given $A = 2^a 3^b$, not A for a given m.	P
32	Halberstam–Richert; Friedlander–Iwaniec, *Opera de Cribro* Ch. 6	Fundamental lemma; input to Theorem A. Explicit form unconfirmed.	U
33	Ding–Li–Li arXiv:2607.03662; Radomskii arXiv:2609.38207 and 2504.09954; Ding–Zhai arXiv:2401.15892; Han–Yu arXiv:2605.15758	Romanoff-type additive results ($n = p + 2^u 3^v$ and relatives), positive proportion. Same ceiling as Erdős–Odlyzko.	P (first two)
34	Sun–Fang, Bull. Austral. Math. Soc. 78 (2008); Chen, Proc. AMS 129 (2001)	One-base extensions of Erdős–Odlyzko; covering iff density zero in a progression.	P

3.7 Corrections found while verifying citations

- arXiv:math/0411144 is by Lettl and Sun (the ledger entry L4 credits Sun alone).
- arXiv:2212.01299 is by Klein, Koukoulopoulos and Lemieux (L5 recorded the authors as not shown).
- Cremona–Koymans is at v3 (12 Oct 2026); Li–Wang–Yi at v2 (Aug 2026).
- Pehlivan's page gives a DOI only, so the journal name stays out of any bibliography until it is checked.

- The JNT 2024 paper on Erdős covering systems in global function fields (ScienceDirect S0022314X24001707) could not be fetched; only its title is known, from a search listing.

4 Methods that have not been used on #203

"Not used" means absent from the repository's record and absent from the sources above. For items 2 and 6 the supporting search was a single query each, so absence there is weak evidence.

#	Method	What it would do	Cost and caveat
1	Function-field analogue: $P^kQ^\ell m+1$ irreducible in $\mathbb{F}_q[T]$ with P, Q coprime primes	Weil-type estimates and exact Chebotarev make both directions testable. Silverman's theorem (math/0401356, abstract level) says $\deg\gcd(a^n-1, b^n-1) \geq cn$ infinitely often there, so the analogue of $c(e)$ can be large, and a covering may exist in that setting. A covering found there would show which mechanism a covering over $\mathbb{Z}$ needs; a proof of non-existence there would show which step of the tail argument survives.	Moderate. CONJECTURED idea; the function-field covering paper above is the entry point.
2	Maynard–Tao weights on the exponent box, in the variable m	For a fixed admissible set of cells the forms $A_i m+1$ are linear in m, so multidimensional sieve weights apply. The conclusion holds on a set of m of density about $(\log x)^{-r}$, which does not reach a positive-density statement.	Low expected gain; listed so the question is closed.
3	Two-dimensional Crittenden–Vanden Eynden principle	A k-coset periodic cover of a lattice line is decided by its first 2^k points. A planar analogue for cyclic-quotient cosets would turn a finite box into a statement about all of $\mathbb{Z}^2$ and convert the search data into a rigorous bound on k.	Unknown; needs a proof.
4	Corvaja–Zannier / Hernandez–Luca for lattice alignments	Controls primes whose lattices share two independent vectors, which is the alignment quantity $c^<$ in `DISTORTION.md`.	Ineffective; finitely many unlisted exceptions.
5	Klein–Koukoulopoulos–Lemieux multiplicity theorem on a lattice line	The direction lemma (Y7) gives a 1-D covering with moduli e_p in the unit-direction case. Whether the multiplicity theorem was then applied is not recorded.	Cheap to check in `C_allE.md`.
6	Reformulation: m is the $\{2,3\}$-free part of $p-1$	NO is the statement that this map from primes with $6 \mid p-1$ is onto the integers coprime to 6. Results on smooth parts of shifted primes may apply.	One literature search not yet made.
7	Exact LP/ILP duals at fixed torus size	Would sharpen Y1 for individual N, but Y1 already excludes every N with all $e_p \leq 4 \cdot 10^5$.	Moot.

Items already used and not repeated: the distortion tilt (Y1, Y5), the Lettl–Sun pruning (Y6, weak), the Li–Wang–Yi framework, the direction-lemma dichotomy, the Romanoff-type lemma for $\{2,3\}$, the fundamental-lemma almost-prime count (N3).

5 Open items carried forward

1. Exact-rational B' for the final δ (resumed today, checkpoint `verdict3/C_ckpt_final.txt`) and the cap-0.28 check (resumed, `verdict3/V_ckpt_cap028.txt`). Theorem Z becomes PROVED when they pass.
2. Exact $B(8 \cdot 10^5)$ (`verdict3/run_verify.sh`, not started today) and Theorem E at $8 \cdot 10^5$.
3. *Opera de Cribro*, Lemma 6.3 or Corollary 6.10: needs the book, possibly through Ashoka's library.
4. Interval arithmetic on Theorem A's floating sums; independent check of `D_eprimes.c`.
5. Prior-art search for Theorem A; one search on item 6 above.
6. The #203 paper (.tex and PDF), with the CRT augmentation lemma added to the pure theorem.